

Terraform - 16 hours

Learn about the infrastructure provisioning using opensource Terraform. This course aims to teach participants on the concept of provisioning infrastructure in an easy, efficient way using Terraform with AWS

After completing this course, candidates will have knowledge of:

- Overview of Terraform and comparison with other tools.
- Setting up and configuring Terraform in Linux
- Working with Terraform CLI commands.
- Understanding Terraform files, variables, providers and provisioners.
- State management using Terraform.
- AWS Basics
- Working with AWS VPC, EC2, ELB, Autoscaling and IAM
- Provisioning AWS Virtual Private Cloud using Terraform.
- Configuring Auto Scaling, Load Balancing using Terraform
- Configuring IAM service with Terraform
- Provisioning and Configuring Containers and Kubernetes using Terraform.
- Getting familiar with Terraform Modules
- Overview and Features of Terraform Enterprise
- GitOps Integration with Terraform

Who Should Attend:

- Developers
- Cloud Engineers
- DevOps Engineers

Day 1: Terraform Fundamentals & AWS Foundations (4 Hours)

Module 1: Prerequisites and Environment Setup (1 Hour)

- Infrastructure as Code (IaC) Concepts
- Terraform Architecture Overview
- Terraform Installation and Configuration
- AWS Account and IAM Setup
- AWS CLI Configuration
- Introduction to HashiCorp Configuration Language (HCL)

Hands-on



- Install Terraform
- Configure AWS CLI
- Verify Terraform Connectivity with AWS

Module 2: Terraform Fundamentals (3 Hours)

- Terraform Workflow
- Providers and Provider Configuration
- Resources and Data Sources
- Variables, Outputs and Locals
- Terraform State File Basics
- Terraform Commands
 - init
 - validate
 - fmt
 - plan
 - apply
 - destroy
- Resource Dependencies

Hands-on: Deploying Basic AWS Resources

- Create an S3 Bucket
- Create IAM Users
- Deploy Security Groups
- Launch an EC2 Instance

Day 2: AWS Infrastructure Provisioning using Terraform (4 Hours)

Module 3: Deploying Networking and Compute Resources (2 Hours)

- Deploying a Custom VPC
- Deploying Public and Private Subnets
- Deploying Internet Gateway
- Deploying Route Tables and Associations
- Deploying Security Groups
- Deploying Amazon EC2 Linux Instances
- Deploying Elastic IP Addresses



Hands-on

- Provision a Complete Network Architecture
- Deploy EC2 Instance into the Custom VPC

Module 4: Deploying Core AWS Services (2 Hours)

- Deploying IAM Users, Groups and Policies
- Deploying Amazon S3 Buckets with Versioning
- Deploying Amazon RDS Instances
- Deploying Amazon DynamoDB Tables
- Using Data Sources to Reference Existing Resources

Hands-on

- Deploy IAM Resources
- Deploy S3 Storage
- Deploy DynamoDB Table

Day 3: Advanced Terraform, Containers & Kubernetes (4 Hours)

Module 5: Advanced Terraform Concepts (2 Hours)

- Remote State Management using Amazon S3
- State Locking using DynamoDB
- Terraform Workspaces
- Terraform Modules
- Reusable Infrastructure Design
- Terraform Module Registry

Hands-on

- Deploy Multi-Environment Infrastructure
- Build and Use Custom Modules

Module 6: Containers and Kubernetes Fundamentals (2 Hours)

- Container Concepts
- Docker Architecture
- Building Docker Images
- Running Containers
- Kubernetes Architecture
- Pods, Deployments and Services



- Introduction to Container Registries

Hands-on

- Build and Run Containers
- Deploy Applications to Kubernetes

Day 4: Modern AWS Services, Security, GitOps & AI Infrastructure (4 Hours)

Module 7: Deploying Container and Serverless Infrastructure (1 Hour)

- Deploying Amazon ECS Clusters
- Deploying AWS Fargate Services
- Introduction to Amazon EKS
- Terraform Support for ECS and EKS

Demonstration

- Deploy ECS/Fargate Infrastructure using Terraform

Module 8: Terraform Security, Enterprise Features & GitOps (1.5 Hours)

- Terraform Security Best Practices
- Managing Secrets and Sensitive Variables
- AWS Secrets Manager Integration
- GitOps Concepts
- GitHub Integration with Terraform
- CI/CD Pipelines for Infrastructure Deployment

Hands-on

- Deploy Infrastructure through GitHub Actions

Module 9: Generative AI Infrastructure using Terraform (1.5 Hours)

- Introduction to Amazon Bedrock
- Deploying Amazon Bedrock Knowledge Bases
- Deploying Amazon Bedrock Agents
- Creating SageMaker Notebook Instances
- Deploying Amazon OpenSearch Serverless for RAG Workloads
- AgentCore Agent Overview
- Managing AI Infrastructure Lifecycle using Terraform

Demonstration



- Deploy Bedrock Knowledge Base Infrastructure
- Deploy AI Supporting Resources using Terraform

Lab and Participants Prerequisites

- Administrator access on the laptop/desktop (64 bit Machine)
- Guest Operating Systems – Preferably minimum Windows 10.
- Access to internet
- Access to SSH protocols for accessing Remote Machines
- Putty and Puttygen tools on participants systems (If participants are using Windows OS). Screenshots have been provided in the 2nd page for installation. SSH client on Linux /Mac machines
- All the labs will be done on AWS Cloud Account
- **Participants must have good hands-on knowledge of Linux, Networking, vi/vim editor tools, AWS Core Services**
- Ports to be opened on participants laptops/ N/W – 22, 3389, 80, 443, 8080, 8443, 3306, 5985, 8140, 3000, 8443
- Open Internet must be allowed to play course and technical videos.